Better Movement Better Life

MEDICOS MINDS

Study Smart Move Better Live Healthy

THE SCIENCE OF MOVEMENT

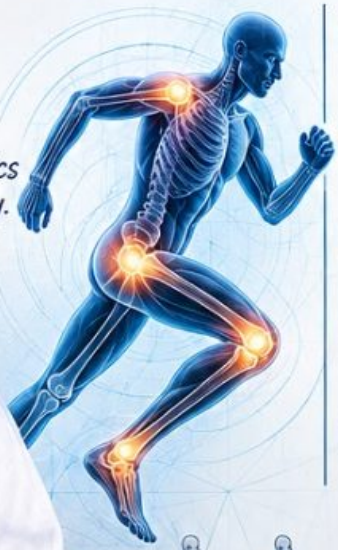
BIOMECHANICS

— MECHANICS OF HUMAN BODY FOR HEALTHCARE PROFESSIONALS —

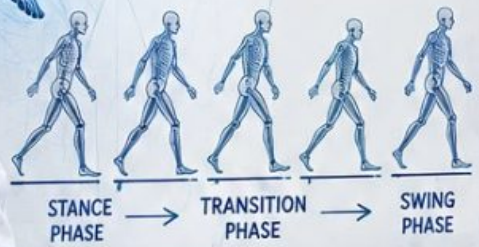
Understand Movement. Improve Function. Enhance Life.



"The body moves in patterns. Biomechanics explains why.



- JOINT KINEMATICS & KINETICS
- MUSCLE FUNCTION & FORCE
- GAIT ANALYSIS & HUMAN MOVEMENT
- POSTURE & ALIGNMENT
- BIOMECHANICAL PRINCIPLES



THIS NOTE IS VALUABLE FOR

MBBS | BHMS | BDS | BPT | BSC.NURSING

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BIOMECHANICS | MEDICOS MINDS

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HIGH-YIELD BPT STUDY GUIDE & EXAM COMPANION

Biomechanics

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OFFICIAL DIGITAL STUDY GUIDE VERIFICATION

Format: High-Yield Digital PDF | Standard Price: Rs. 99

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STUDY GUIDE DETAILS & CONTENTS:

Comprehensive high-yield notes designed for Bachelor of Physiotherapy (BPT), MBBS, BDS, and allied healthcare students. Covers fundamental anatomy, clinical assessments, pathological conditions, evidence-based physical therapy rehabilitation protocols, and exam-oriented high-yield review.

KEY HIGHLIGHTS:

- Exam-Focused: Concise points curated from leading medical textbooks and clinical references.
- Visual Diagrams: High-yield diagrams and tables for quick revision.
- Mobile & Tablet Optimized: Readable on phones, tablets, iPads, and laptops.
- Official Medicos Minds Release: Verified by Dr. MD Kamran Alam.

8. JOINT STPUCTURE AND FUNCTION

A joint (articulation) is the meeting point of two or more bones or cartilages. It provides stability and allows movement.

1. Basic Structure of a Synovial Joint :-

1. Articular Cartilage – Covers the ends of bones, reduces friction and absorbs shock.
2. Joint Capsule – Encloses the joint.
   - Outer fibrous layer – provides strength.
   - Inner synovial membrane – secretes synovial fluid.
3. Synovial Fluid – Lubricates the joint and nourishes the articular cartilage.
4. Ligaments – Connect bone to bone and provide stability.
5. Muscles and Tendons – Produce movement and support the joint.
6. Bursae – Fluid filled sacs which reduce friction between tendons/muscles and bones.
7. Menisci / Articular Discs – Fibrocartilaginous pads which improve congruency and absorb shock.

2. Classification of Joints :-

A. Structural (By Structure)
- Fibrous Joints – No joint cavity. (e.g. Sutures of skull)
- Cartilaginous Joints – Cartilage present. (e.g. Symphysis pubis)
- Synovial Joints – Joint cavity present. (e.g. Knee, Shoulder)

B. Functional (By Movement)
- Synarthrosis – Immovable joint. (e.g. Skull sutures)
- Amphiarthrosis – Slightly movable. (e.g. Intervertebral joint)
- Diarthrosis – Freely movable. (e.g. Shoulder joint)

4. Functions of Joints :-

- Provide support and stability to the body.
- Allow movement and mobility.
- Transmit forces from one part to another.
- Absorb shock and reduce friction.
- Play an important role in posture and balance.

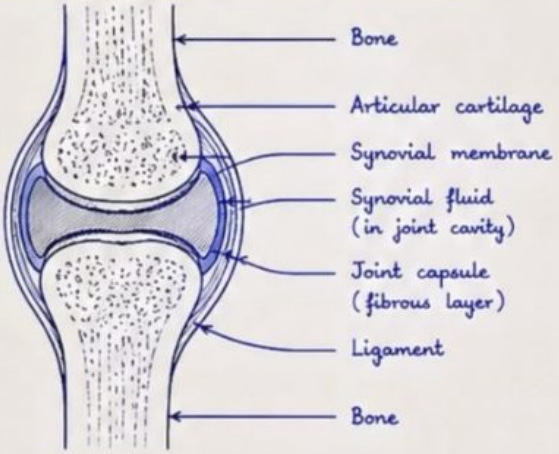
5. Movements at a Synovial Joint :-

- Flexion – Decreasing the angle.
- Extension – Increasing the angle.
- Abduction – Moving away from midline.
- Adduction – Moving towards midline.
- Medial Rotation – Turning towards midline.
- Lateral Rotation – Turning away from midline.
- Circumduction – Cone shaped movement.
- Pronation – Turning palm backward.
- Supination – Turning palm forward.
- Dorsiflexion – Upward movement of foot.
- Plantarflexion – Downward movement of foot.

6. Importance in Physiotherapy :-

- Knowledge of joint structure helps in assessment.
- Helps in planning effective treatment.
- Prevents injuries by improving joint stability.
- Useful in mobilization, strengthening and rehabilitation.

Structure of a Synovial Joint (e.g. Knee Joint) :-



3. Types of Synovial Joints (By Shape) :-

1. Plane (Gliding) – Allows gliding. e.g. Intercarpal joints
2. Hinge – Allows flexion & extension in one plane. e.g. Elbow joint
3. Pivot – Allows rotation around a central axis. e.g. Atlanto-axial joint
4. Condyloid (Ellipsoid) – Allows flexion, extension, abduction, adduction & circumduction. e.g. Wrist joint
5. Saddle – Allows flexion, extension, abduction, adduction & some rotation. e.g. 1st CMC joint of thumb
6. Ball and Socket – Allows movement in all directions including rotation. e.g. Shoulder, Hip joint

Quick Note :-
- → Joints are classified by structure, function and shape.
- → Synovial joints are freely movable.
- → Stability comes from capsule, ligaments, muscles and proper alignment.
- → Movement is produced by muscles acting on joints.

THANK YOU FOR YOUR PURCHASE!

We wish you the very best in your medical and physiotherapy examinations.

For any questions, supplementary questions, or study support, reach out to us:

SUPPORT & COMMUNITY:

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